

Read this first

Ask before you explain. Explain only after they've tried.

Keep it warm

Insulation is trapped air

Say	"Two bottles, same warm water. One gets a scarf. Which one wins?"
Ask first	"Which material will hold the heat best: the scarf, the newspaper or the leaves?"
After: explain	Heat always moves from warm things to cold things. A scarf does not make heat; it traps a layer of still air, and still air is very slow at carrying heat away. That is why a loose, fluffy wrap beats a thin tight one.
Look around	Snow does the same job: a layer of snow traps air and keeps the ground under it warmer than the air above.
Close with	"How would you wrap it if you had to keep it warm all day?"

The slowest falling leaf

Air resistance and autorotation

Say	"Find me the leaf that takes the longest to reach the ground."
Ask first	"Wide and flat, or narrow and curled: which falls slower?"
After: explain	A wide, flat leaf pushes against more air, so it falls slowly. A maple or sycamore seed does something cleverer: its wing is angled, so falling air makes it spin, and the spinning wing sweeps even more air. It is the same trick as our paper helicopter.
Close with	"Could you fold a leaf so it falls even slower?"

Breath clouds

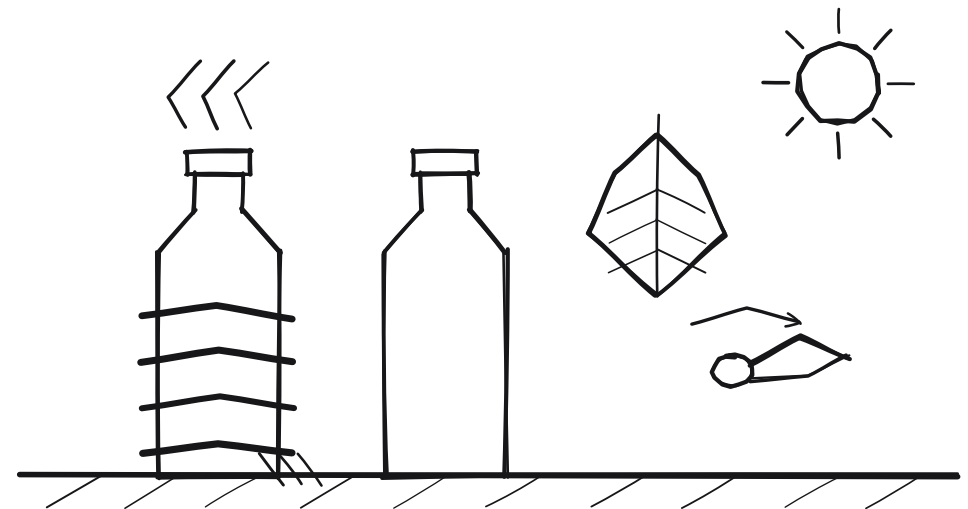
Water vapour condensing

Say	"Make a cloud with your mouth."
Ask first	"Why can we see it out here but not in the kitchen?"
After: explain	Your breath is warm and full of invisible water vapour. Outside, the cold air cools it quickly and the vapour turns into tiny floating droplets, which is what you can see. Indoors, the air is warm enough to keep the water invisible.
Look around	The same thing happens on a cold car window, on a cold drink in summer, and behind a plane.
Close with	"How cold do you think it has to be before you can see your breath?"

Twenty minutes in the cold is enough. Warm tap water only, never boiling. Never walk on a frozen pond or canal, however solid it looks.

Cold outside

3 challenges for 20 minutes in the cold, ages 8 to 13



The cold is the material. Wrap up and go out.

Grown-ups: start on the back page.



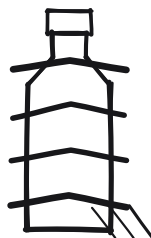
Take these outside with you

Everything fits in one bag.

2 identical bottles with caps	1 scarf, jumper or old towel	1 newspaper or dry leaves
1 warm (not hot) water from the tap	1 stick to mark the ground	1 pair of thin gloves

How it works

- 1 Fill both bottles with the same warm water before you leave.
- 2 Start challenge 1, then play the other two while you wait for it.
- 3 Twenty minutes outside is plenty. Go back in while everyone still wants more.



Our results

Date _____ How cold was it? _____

Warmest bottle was wrapped in

Slowest leaf

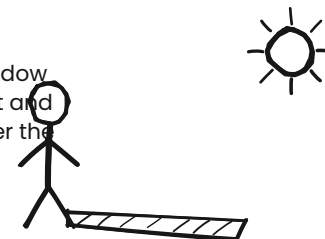
Longest breath cloud

My shadow was ____ steps long

Engineers _____

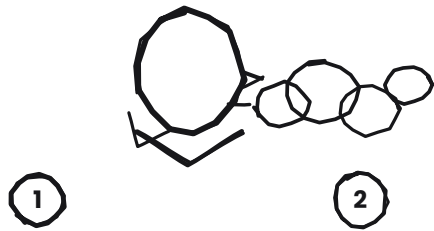
Take home: how long is your shadow?

At midday, stand still and mark the end of your shadow with a stick. Count your own steps back to your feet and write the number down. Do it again in June: in winter the sun stays low, so the shadow is much longer.



Breath clouds

5 minutes



1

Breathe out slowly and watch the cloud

2

Whose cloud lasts longest?

3

Try it again inside the door

Four more things to try

1

Breathe out through your nose, then through your mouth. Guess first: which cloud is bigger?

2

At dusk, shine a torch through your breath. It suddenly shows up much brighter.

3

Breathe on a cold window until it fogs, then write your name in it.

4

Breathe through a straw. Who can send their cloud the furthest?

Did you know

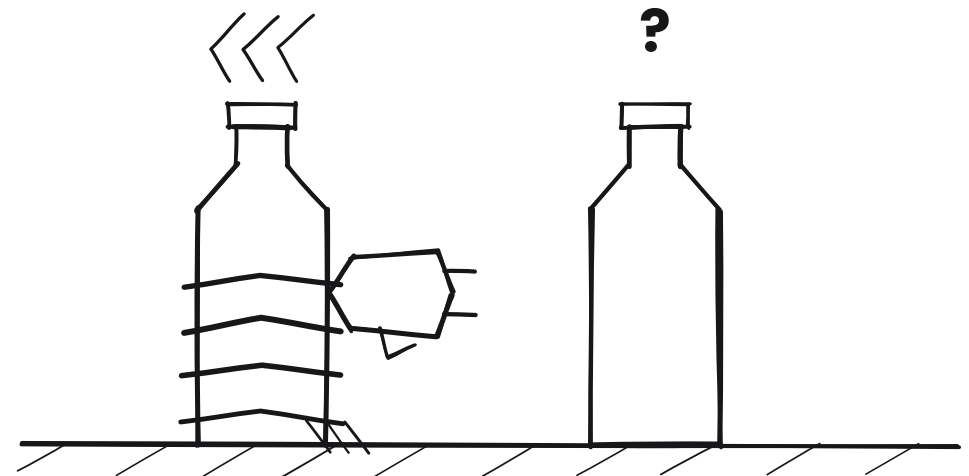
The white trails behind aeroplanes are the same thing: hot, damp exhaust meets freezing air and turns into tiny droplets and ice crystals. Real clouds form this way too, when damp air rises and cools.

Levels

- Make a cloud you can see ☐
- Make the longest-lasting one ☐
- Guess how cold it has to be ☐

Keep it warm for 20 minutes

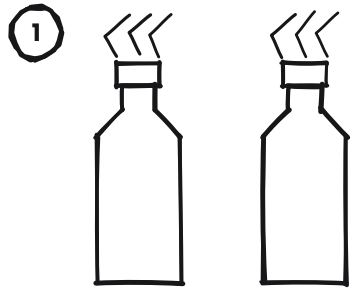
15 minutes



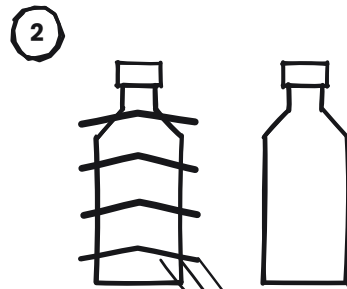
Which bottle will still be warm?

Levels

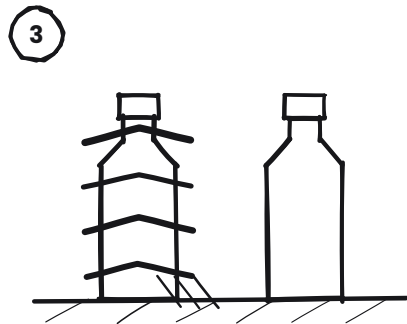
- The wrapped bottle stays warmer ☐
- Rank three materials from best to worst ☐
- Predict the order before you test it ☐



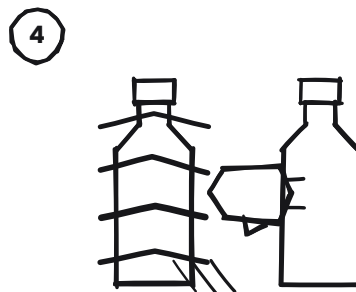
Same warm water in both bottles



Wrap one, leave the other bare



Leave them side by side outside



After 20 minutes, feel both

The slowest falling leaf

10 minutes



Collect leaves of different shapes



Drop two at a time from the same height



Find a spinning seed and race it too

Levels

■□□ Find the slowest leaf



■■□ Beat it with a folded one



■■■ Explain why the seed spins

